



Financial Institution and Market



CHAPTER 1:Overviews of Financial System

1.1 Introduction



- What is Finance ??

1.1. Introduction

What is Finance

- *Finance is the application of economic principles to decision-making that involves the **allocation of money** under conditions of uncertainty.*
- Investors allocate their funds among financial assets in order to accomplish their objectives, and businesses and governments raise funds by issuing claims against themselves that are invested.
- Finance provides the framework for making decisions as to how those funds should be obtained and then invested.

Cont....

- It is generally agreed that the field of finance has three specialty areas:
 1. **Capital markets and capital market theory:** The specialty field of capital markets and capital market theory focuses on the study of *the financial system, the structure of interest rates, and the pricing of risky assets.*
 2. **Financial management:** is the specialty area of finance concerned *with financial decision-making* within a business entity.
 3. **Investment management:** is the specialty area within finance dealing with the *management of individual or institutional funds.*

1.2 The Financial System



- **What is Financial System?**

- Financial system is platform by which funds are transferred from those entities that have funds to invest to those entities that need funds to invest.
- The financial system of an economy consists of three components:
 - (1) financial markets;
 - (2) financial intermediaries; and
 - (3) financial regulators.
- For this reason, we refer to this specialty area as **financial markets and institutions.**

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- It is through a country's financial system that entities with funds *allocate those funds* to those who have potentially more productive ways to deploy those funds, potentially leading to faster growth for a country's economy.
- A financial system makes possible a **more efficient transfer of funds** is by overcoming the information asymmetry problem between those with funds to invest and those needing funds. In general, *information asymmetry means* that one party to a transaction has more or superior information than the other party, resulting in an imbalance of power in a transaction.
- In terms of a financial system, information asymmetry can lead to an inefficient allocation of financial resources.

1.3 Market

- It is an actual or nominal place where forces of demand and supply operates and where buyer and seller interacts (directly or through intermediaries)to trade products.

1.3.1Types of market

□ Market can be classified on different basis. Such as:

A. On basis of geographical are

- i. Local market: is limited in a certain place
- ii. Regional market : are not limited in a certain place but expanded in a regional level
- iii. National market :if buying and selling of some product is done in the whole nation.
- iv. International market



B. On basis of control

- I. Regulated market
- II. Unregulated market

C. On basis of consumption

- i. Consumption market : Consumable Product
- ii. Industrial market : Raw Material

D. On the basis of competition

- i. Monopoly market
- ii. Perfect market
- iii. Imperfect market



E. On basis of volume of business

- i. Wholesale market
- ii. Retail market

F. On basis of nature of product

- i. **Commodity market**

The market where consumer and industrial commodity like clothes, rice, machines and equipment are bought or sold.



ii. Financial market

A financial market is a market where financial instruments are exchanged.

1.4 Financial markets

- A financial market is a market where financial instruments are exchanged.
- The more popular term used for the exchanging of financial instruments is that they are “traded.”
- Financial markets provide the following three major economic functions:
 - Price discovery
 - Liquidity
 - Reduced transaction costs

1.4.1 Role of Financial Markets

1. Price Discovery

- It determine price or required rate of return of asset.
- *Price discovery means that the interactions of buyers and sellers in a financial market determine the price of the traded asset.*
- Equivalently, they determine the required return that participants in a financial market demand in order to buy a financial instrument.

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- Because the motivation for those seeking funds depends on the required return that investors demand it is. This function of financial markets that signals how the funds available from those who want to lend or invest funds will be allocated among those needing funds and raise those funds by issuing financial instruments.

2. Provide liquidity.

- Financial markets provide a opportunity for **investors to sell a financial instrument** and is said to offer **investors “liquidity.”**
- This is an attractive feature when circumstances arise that either force or motivate an investor to **sell a financial instrument.**

- **Without liquidity**, an investor would be **forced** to **hold** onto a financial instrument until either conditions arise that allow for the **disposal of the financial instrument** or the issuer is contractually obligated to **pay it off**.
- **For a debt instrument**, that is when it matures, whereas
- **For an equity instrument** that is until the company is either **voluntarily or involuntarily liquidated**

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- All financial markets provide some form of liquidity. However, the degree of liquidity is one of the factors that characterize different financial markets.

3. Reduce transactions costs:

Efficiency

- It reduces the cost of transacting when parties want to trade a financial instrument.
- In general, one can classify the costs associated with transacting into two types:
- **search costs and information costs.**
- *Search costs in turn fall into categories:*
- explicit costs and implicit costs.

Others

4. *Borrowing and Lending*: Financial markets permit the transfer of funds (purchasing power) from one agent to another for either investment or consumption purposes.

5. *Risk Sharing*: Financial markets allow **a transfer of risk** from those who undertake investments to those who provide funds for those investments



6. *Information Aggregation and Coordination*: Financial markets act as collectors and aggregators of information about **financial asset values and the flow of funds from lenders to borrowers**



Unit Two:

Types Financial Market: Financial asset

2.1 Definitions Of Financial Market

- **Financial market** is a marketplace, where creation and trading of **financial asset**.
- It plays a crucial role in allocating limited resources, in the country economy.
- It act as intermediary between the saver and investor by mobilising funds b/n them.
- What does mean financial asset?

2.1.1 what is a financial asset ?

- A claim against income or wealth of the individual, business firm and government

2.1.1.1 Characteristics of Financial Asset

- It usually represent by certificate, computer record file, or legal document
- Do not depreciate like physical goods
- Have a little or no value as a commodity
- Cost transportation or storage are low
- They are fungible

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- Usually related to the lending money
 - Physical condition or form usually not relevant in determining market value
 - FA sought after are b/c they promise future returns to their owners and serve as a store of value (purchasing power).

Roles of financial asset



- Transfer funds from surplus units to deficit units.
- Transfer funds so as to redistribute unavoidable risk associated with cash flows generated from both tangible and intangible assets.

2.1.1.2 How Financial asset are created ?

- Internal financing to acquire asset ?
 - ✓ use current income
 - ✓ use accumulated saving
- External financing to acquire assets
 - ✓ raise funds by issuing liabilities(debts)
 - ✓ raise funds by issuing stock (equities)

2.1.1.2 Classifications of financial market

- By the **maturity of claim** can be categorise in to two. Such as :
 1. Money market
 2. Capital market

1. Based on maturity :Money market

A. Money Market

- The money market is the sector of the financial market that includes financial instruments that have a maturity or redemption date that is one year or less at the time of issuance.

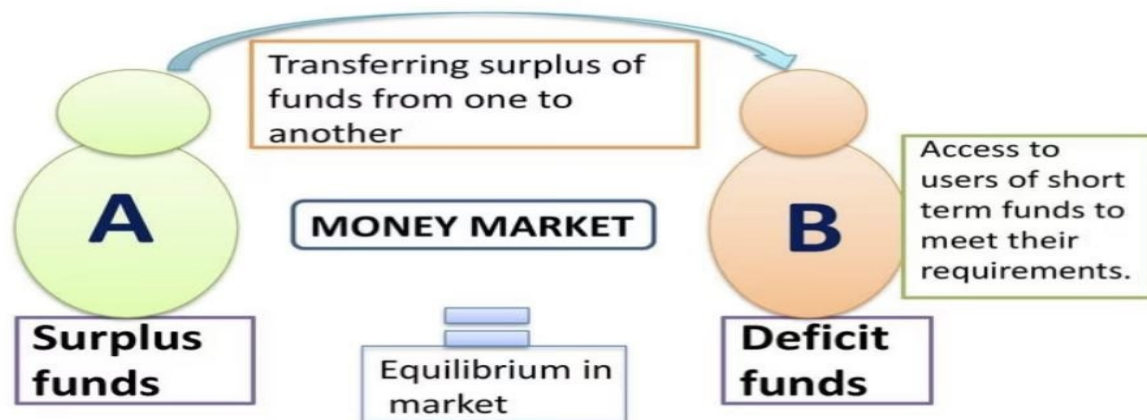
Characteristics of Money Market

- The money market securities have three basic common characteristics:
 1. They are usually sold in **large denominations**. Money Market Mutual funds can mitigate (alleviate) this problem
 2. They have **low default risk**
 3. **They mature in one year or less from their original issue date**. Most money market instruments mature in less than 120 days
- Money market transactions do not take place in one particular location or building rather traders usually arrange purchase and sales between participants over a phone and complete them electronically.
- Because of this money market securities usually have an active secondary market.

- Money markets are wholesale markets: most transactions are very large as a result this large size most individual investors cannot invest their money directly in the money markets.
- To solve this problem dealers and brokers, operating in trading rooms of large banks and brokerage houses, bring customers together.

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- Despite the wholesale nature of the market, innovative securities and trading methods help investors enjoy the benefits of money market securities.

Objectives of Money market



Following are some of the important objectives of a money market:

1. Provides access to users of short term funds to meet their requirements at reasonable cost.
2. Helps in transferring surplus from one sector to another.
3. Provides a mechanism for maintaining equilibrium in the short term surplus and deficits of funds in the market.

Participants of the Money Market

- An obvious way to discuss the participants of the money market would be to list those **who lend and borrow in the money market**.
- However, this approach do not work here because, money market participants involve in both sides of the market.
- For instance, a bank may borrow huge amount of funds via Certificates of Deposits (CDs), and at the same time it will advance short-term loans to its clients.
- Nevertheless, in the following sections we will discuss the principal players of the money market.

Central banks

- **Central Banks:**
- conduct monetary policy by adjusting the supply of money, usually through buying or selling securities in the open market.
- are frequently engaged in injection and absorption of money supply in the economy using a tool known as open market operations.
- The central bank's responsibility for **money supply** makes them the single most influential participant in the money market.

Commercial banks

- CBs hold a large percentages of government securities than any other institutions in many countries including Ethiopia.
- Commercial banks in Ethiopia are the dominant players in the country's money market.
- In many countries regulations prohibit CBs from investing in high risk corporate bonds and stocks that have a higher rate of returns but entail huge associated risks.
- As money markets instruments provide an almost risk less investment opportunity with an added advantage of high liquidity, the CBs are major issuers of money market instruments.

Business

- Many business firms buy and sell securities in the money market to finance their short-term capital needs as well as to warehouse their excess funds.
- Many large corporations and firms engage in both sides of the market as money market investments are substantial in size.

Investment and Securities Firms

1. **Investment Companies**: Investment companies sell shares to the public and invest the proceeds in a diversified portfolio of securities.

2. **Investment Banking Firms**

They serve two functions:

- a. For entities that need funds, they assist in obtaining those funds.
- b. For investors who wish to invest funds, they act as brokers or dealers in the buying and selling of securities.

They are large diversified **brokerage firms** that are active in the market to trade on behalf of their commercial accounts.

- They are very important to the liquidity of the money market because they ensure that both buyers and sellers can readily market their securities.



Roles (services) of Investment Banks

1. Research
2. Sales and Trading
3. Raising Capital
4. Underwriting
5. Merger and Acquisition
6. Restructuring

1. Research



- Larger investment banks have large teams that gather information about companies and offer recommendations on buying or selling their stock. They may use these reports internally but can also generate revenue by selling them to hedge funds .

2. Sales and Trading

- This sales and trading activity in investment bank is much different from what you will find in research.
- Investment banks deals with sales and trading of securities in **secondary market**.

3. Raising Capital



Investment banks help their clients for raising capital through IPO and private placements.

- Client's takes help of investment banks for capital raising.

4. Underwriting

- An underwriter can work for financial institutions such as mortgage companies, stock exchanges, insurance companies, and banks.
- They are mainly interested in gathering all available information regarding the financial standing of the companies, precisely estimating the risk involved, and assisting them in deciding whether or not to accept a new contract.

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- Underwriters play a crucial role in assessing and evaluating risk for various financial organisations such as mortgage, insurance, loan, or investment companies.
 - They use their specialised knowledge and expertise to decide whether a contract is worth the risk.

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- The specific information an underwriter assesses is subject to the case they are reviewing.
 - As an example, a health insurance company's underwriter examines the health risks associated with applicants, whereas a loan underwriter evaluates aspects such as credit history.

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- Investment banks provide underwriting of securities and broking related services for their clients.
 - Investment banks plays intermediary role between his client and the investors.

5. Merger and Acquisition

This is one of the major functions which are performing by an investment bank.

- Investment banks provide advisory regarding to M&A to its clients as they can create different M&A models.
- Investment bankers do growth or dilution analysis to analyze company.
- Also Banks check synergies analysis.

5.1 Tasks of M&A

1. **Providing industry overviews:**

What are the current dynamics in an industry?

For example, in the oil and gas industry, it's all about the energy transition. In the retail industry, it's all about the scramble for e-commerce. Providing a detailed industry overview is the job of investment banks, and informs the decision about which companies are most relevant for the M&A process.



2. **Deal origination:** Investment bankers first find companies that are suitable and attractive, and then narrow this down to companies that are possible to acquire or merge with.



3. Company Valuation: Valuing companies, public and private, is the bread and butter of investment banks. An investment bank will usually develop a valuation using a complex financial model that takes account of all of the target company's current and projected financial results.



4. Negotiation: CEOs can't easily approach other CEOs with some form of intermediary. When an investment banker is the intermediary, the intention for everyone is obvious: one side is looking for a transaction of some form, and wants to be transparent about it.

- If this leads to some form of negotiation, the investment banker will be brought on board to help their client in the process.



5. Due Diligence: the fact that experienced investment bankers will have participated as intermediaries on dozens of deals makes them far more capable of knowing the issues to look out for at the due diligence phase than the average business owner.

- The investment bank will also have their own legal team to look over regulatory and compliance issues relevant to the deal.



6. Deal Closing: Deal closing is not as easy as signing on a dotted line and walking away with the ownership of a company. Rather, it involves a careful final assessment of all the terms and conditions involved, and accounting for contingencies. Again, given that investment bankers will have done this dozens of times, they're far more capable than the standard business owner of ensuring that nothing is missed in the closing.

6. Restructuring

- What do you mean by restructuring?
- An action taken by a company to drastically change the financial and operational features of the company is known as restructuring. This is typically done when the business is undergoing financial problems. It includes modifying the debt, operations, or structure of a company to limit financial harm

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- ❑ Investment Bank plays very important role in restructuring and reorganizing of capital.
 - ❑ Investment banking performs restructuring of a company for the purpose of making it more profitable.

Investment bank Vs Commercial bank

Commercial Banks:

- Manage deposit accounts
- Make loans to the public using the money held on deposit.
- Highly Regulated and hence are less tolerance of risk

Investment Banks:

- Facilitate the buying and selling of stocks, bonds and other investments.
- Help companies to go public initial offerings (IPO)
- Loosely regulated and hence have high risk tolerance

Investment and Secu...

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3. **Finance Companies:** lend funds to individuals.
4. **Insurance Companies:** property and casualty insurance companies must maintain the liquidity because of their unpredictable need for funds.
 - Such firms, therefore, should invest some amount of their investment funds in money market securities to raise cash during emergency situations.
- 5.

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- **Pension Funds:** these companies invest a portion of their cash in money markets so that they can take advantage of investment opportunities that they may identify in the stock or bond markets.

Individuals

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- With the special help of the **money market mutual funds** individual investors with relatively small amounts of cash to invest can get an access to invest on large denomination money market instruments.

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- Typically, money market instruments are debt instruments and include Treasury bills, commercial paper, negotiable certificates of deposit, and repurchase agreements.

a. Treasury Bills

- *Treasury bills (popularly referred to as T-bills) are short-term securities issued by the government; they have original maturities of either four weeks, three months, or six months.*
- T-bills carry no stated interest rate.

- Instead, they are sold on a *discounted basis*.
- *This means that the* holder of a T-bill realizes a return by buying these securities for less than their maturity value and then receiving the maturity value at maturity

***b.* Commercial Paper**

- *Commercial paper is a promissory note—a written promise to pay—* issued by a large, creditworthy corporation or a municipality.
- This financial instrument has an original maturity that typically ranges from one day to 270 days.

- Most commercial paper is backed by bank lines of credit, which means that a bank is standing by ready to pay the obligation if the issuer is unable to due.
- Commercial paper may be either interest bearing or sold on a discounted basis

c. Certificates of Deposit

- *Certificates of deposit (CDs) are written promises by a bank to pay a depositor.*
- Nowadays, they have original maturities from six months to three years. *Negotiable certificates of deposit are CDs issued by large commercial banks that can be bought and sold among investors.*
- On the maturity date, the investor is repaid the principal plus interest.

d. Euro dollar

- A *Eurodollar CD* is a negotiable CD for a U.S. dollar deposit at a bank located outside the United States or in U.S. International Banking Facilities.
- The interest rate on Eurodollar CDs is the London Interbank Offered Rate (LIBOR), which is the rate at which major international banks are willing to offer term Eurodollar deposits to each other.

e. Repurchase Agreement

- Another form of short-term borrowing is the repurchase agreement.

- This common lending agreement is referred to as a *repurchase agreement* or *repo* because it specifies that the:
 - (1) *the borrower sell the bonds to the lender* in exchange for proceeds; and
 - (2) *a some specified future date, the borrower repurchases the bonds from the lender at a specified price.*
- The specified price, called the *repurchase price*, is higher than the price at which the bonds are sold because it **embodies the interest cost** that the lender is charging the borrower.

- The interest rate in a repo is called the *repo rate*.
Thus, a repo is nothing more than a collateralized loan.
- *It is* classified as a money market instrument because the term of a repo is typically less than one year.

Based on Maturity :Capital Market

- The *capital market* is the sector of the financial market where *long-term* financial instruments issued by corporations and governments.
- Here “long-term” refers to a financial instrument with an original maturity greater than one year and perpetual securities (those with no maturity).

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- There are two types of capital market securities:
 - A. those that represent shares of ownership interest, also called *equity*, *issued by corporations*, and
 - B. *those that* represent indebtedness, **issued by corporations and issued by the governments.**

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- A capital market debt obligation is a financial instrument whereby the borrower **promises to repay the maturity value one year after issuance.**
 - These debt obligations can be broken into two categories: bank loans and debt securities.

- Debt securities include
 - (1) bonds,
 - (2) medium-term notes, and
 - (3) asset-backed securities.
- The distinction between a bond and a note has to do with the number of years until the obligation matures when the security is originally issued. Historically, a note is a debt security with a maturity at issuance of 10 years or less; a bond is a debt security with a maturity greater than 10 years

- a bond, a note, or a medium-term note could be designated as simply a *bond*.
- *We will refer to the investors in any debt obligation as either the debt holder, bondholder, or note holder.*
- Capital market can be divided into two. Such as primary and secondary market

Primary Market

- When a listed companies first issued and sold the financial instrument in the *primary market*.
- This is the market in which new issues are sold and new capital is raised.
- So it is the market whose sales directly benefit the issuer of the financial instrument.
- Issuance of securities must comply with the securities laws.
- The primary market can be classified as the public market and the private placement market.

Public Market Issuance

- The public market offering of new issues typically involves the use of an investment bank (d/t from country to country). Investment banks involved by *underwriting of securities*.
- *A company offer securities to the **general public** to raise funds to finance its long term goals.*
- *Securities are issued either by an Initial public offer (IPO)*

Private Placement Market

- There are different regulatory requirements for securities that are issued to the **general investing public** and those that **are privately placed**.
- Private placement offerings are securities released for sale only to **accredited investors**
- all securities offered to the general public Usually must be registered with the SEC.

Secondary Market

- A secondary market is one in which financial instruments are resold among investors.
- No new capital is raised and the issuer of the security does not benefit directly from the sale.
- Trading takes place among investors.
- Investors who buy and sell securities on the secondary markets may obtain the services of *stock brokers, individuals who buy or sell securities for their clients.*

Secondary markets for common stock

- The market shows that the opinions of investors about the economic prospect of a company are expressed through the trades they execute.
- These trades together give the market consensus opinion about the price of the stock.

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- Secondary trading of common stock occurs in a number of trading locations:
 - Major national stock exchanges
 - Regional stock exchanges
 - Over the counter market and ,
 - Independent Electronic Trading Systems

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- Stock Exchanges
 - ▣ trading floors
 - Over-The-Counter Market
 - ▣ no trading floor
 - ▣ negotiated system
 - ▣ multiple market maker system
 - Independent Electronic Trading Systems

Major National Stock Exchanges

- In US there are two
 - New York Stock Exchange (NYSE)
largest, about 2300 companies listed.
 - American Stock Exchange (ASE or AMEX)
Second largest in US, about 1100 firms are listed

Regional Exchanges

- On these exchanges there are two kinds of stocks exchanges:
 - ▣ Stocks of companies that **could not qualify for listing** (or **do not wish to list**) on one of the major national exchanges
 - ▣ Stocks that are also listed in one of the major national stock exchanges(which are dually listed stocks).

Over the counter Market

- Is a market for unlisted stocks.
- **The National Association of Securities Dealers (NASD)**, a private organization, represents and regulates **the dealers in the OTC market under the supervision of the SEC.**
- The National Association of Securities Dealers Automatic Quotation (NASDAQ) system is **an electronic quotation system** that provides price quotations to market participants about the more actively traded issues in the OTC market.
- There are about 3,800 common stocks included in the NASDAQ system.

Fourth Market (Independent Electronic Trading Systems)

- It is not always necessary for two trans actors to use the services of a broker or a dealer to execute a transaction.
- **The direct trading of stocks between two transactors(customers)** without the use of a broker is called the fourth market.
- Institutional investors cross trades (match buyers and sellers) via computer.

- Internet based trade networks: e.g. Instinet (the largest)
- Customers can meet directly (no broker)
- Used mostly by professional money managers
- Both (listed and unlisted companies)
- Advantage:
 - fewer intermediaries
 - Transparency
 - Faster Execution
 - After-hours trading
- Disadvantage:
 - less liquidity since Only large blocks of securities are traded (Fewer people to trade with)
- Fastest growing markets

Role of Dealers in Exchange and OTC Markets

- Dealers are market makers.
- On the exchanges there is only one market maker or dealer per stock, known as the **specialist**.
- Specialists realize a profit only from those trades in which they are involved.

2. Based on nature of claim

- It can be categorise in to two. Such as
 1. Equity market
 2. Debt market

2. Based on nature of claim :Equity market

- A. **Equity market** : is a market where in the investor s deals in equity instruments
 - It is the market for residual claims
 - Equity securities represent **an ownership interest** in a corporation.
 - Types of equity securities:
 - Common stock
 - Preferred stock

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- Common stockholders are the residual owners.
 - ▣ Right to income
 - ▣ Right to assets
 - Preferred Stockholders have preference over common stockholders.
 - ▣ Right to fixed dividend

Stock Exchanges

- Stock exchanges are formal organizations that are made up of **members** that use the facilities to exchange certain common stocks and other securities.
- Regulated by the SEC
- Stocks that are listed on an exchange are said to be **listed stocks**.
- To be **listed**, a company must apply and satisfy requirements established by the exchange where listing is sought.

Based on nature of claim :debt market

- Debt market is a place where the short term debt (T-bills, commercial paper, bank deposit certificate and repurchase bills are traded) and long-term debt (bond) are traded in the market.

The Bond Market : Overview of the Bond Markets

- A bond is a **promise** to make periodic coupon payments and to repay principal at maturity; breach of this promise is an event of default, or
- Bond is a long-term debt investment in which an investor loans money to an entity (corporate or governmental) that borrows the funds for a defined period of time at a fixed interest rate.
- carry original maturities greater than one year so bonds are instruments of the capital markets
- issuers are corporations and government units

Basics of Bond

- Bonds are used as a means of financing variety of projects and activities by companies, municipalities, state, local and foreign governments.
- Bonds are commonly referred to as **fixed-income securities** and are one of the three main financial asset classes, along with stocks and cash equivalents.
- The place where bonds and other debt instruments are sold is called the **debt market**.

- Corporate borrowers issue bonds both to raise finance for major projects and to cover ongoing and operational expenses.
- Bonds are also issued by public authorities, credit institutions, companies and supranational(Multinational) institutions in the primary markets.
- The most common process of issuing bonds is through **underwriting**.

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- In underwriting, one or more security firms or banks, forming a syndicate/association, buy an entire issue of bonds from an issuer and re-sell them to investors.
 - The security firm takes the risk of being unable to sell on the issue to end investors.

Types of bond



1. **Treasury Notes and Bonds**
2. **Municipal bond**
3. **Corporate bond**

1. Treasury Notes and Bonds

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- T-notes and T-bonds issued by the U.S. treasury to finance the **national debt and other federal government expenditures**
- Backed by the full faith and credit of the U.S. government and are default risk free
- Pay relatively low rates of interest (yields to maturity)
- Given their longer maturity, not entirely risk free due to interest rate fluctuations
- **Pay coupon interest (semiannually), notes have maturities from 1-10 yrs, bonds 10-30 yrs**

2. Municipal Bonds (munis)

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- Securities issued by state and local governments to fund either temporary imbalances between operating expenditures and receipts or to finance long-term capital outlays for activities such as school construction, public utility construction or transportation systems.
- Tax receipts or revenues generated are the source of repayment
- Attractive to household investors because interest (but not capital gains) are tax exempt

3. Corporate Bonds

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- All long-term bonds issued by corporations
- Minimum denominations publicly traded corporate bonds could be set.
- Generally pay interest semiannually
- Bond indenture
 - ▣ legal contract that specifies the rights and obligations of the bond issuer and the bond holder

3.By organizational structure

1. **Exchange traded market** : a financial market which has a centralised organization with the standardised procedure
2. **Over the counter market** : An OTC is characterised by a decentralised organization having a customised procedure.

4. By time of delivery

1. **Cash market** : the market where the transaction between the buyer and sellers are settled in the real time
2. **Future market/ Derivative market** :is one where the delivery or settlement of commodities takes place at a future specified date

Derivative Market

- Financial markets are classified in terms of cash market and derivative markets.
- The cash market, also referred to as the spot market, is the market for the immediate purchase and sale of a financial instrument.
- In contrast, some financial instruments are contracts that specify that the contract holder has either the obligation or the choice to buy or sell another something at or by some future date. The “something” that is the subject of the contract is called the *underlying*.

Chapter 3: The Derivatives Market



3.1. Introduction

3.2. Basics of Derivative Market

3.3. Types of Basic Derivatives

3.3.1. Introduction to forward Markets

3.3.2. Introduction to Financial Futures Markets

3.3.3. Introduction to Options Markets

3.1 Introduction

- **Definition of Derivatives**

The term “derivatives” is used to refer to financial instruments which **derive their value** from some underlying assets.

- The underlying assets could be equities (shares), debt (bonds, T-bills, and notes), and currencies.

Intro...

- The basic purpose of derivatives is **to transfer the price risk** (inherent in fluctuations of the asset prices) from one party to another; they facilitate the allocation of risk to those who are willing to take it.
- In so doing, derivatives help mitigate the risk arising from the future uncertainty of prices.
- For example, on November 1, 2011 a coffee farmer may wish to sell his harvest at a future date (say January 1, 2012) for a pre-determined fixed price **to eliminate the risk of change in prices** by that date.
- Such a transaction is an example of a derivatives contract.
- The price of this derivative is driven by the spot price of coffee which is the "underlying asset".

3.2. Basics of Derivative Markets

Spot Market

- In the context of securities, the **spot market or cash market is a securities market in which** securities are sold for cash and delivered immediately.
- The delivery happens after the settlement period.

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- Stock prices fluctuate continuously during any given period.
- Prices of some stocks might move up while that of others may move down.
- In such a situation, what can we say about the stock market as a whole?
- Has the market moved up or has it moved down during a given period?
- Similarly, have stocks of a particular sector moved up or down?

Index ...

- To identify the general trend in the market (or any given sector of the market such as banking), it is important to have a reference barometer which can be monitored.
- An index is a basket of identified stocks, and its value is computed by taking the weighted average of the prices of the constituent stocks of the index.
- A market index for example consists of a group of top stocks traded in the market and its value changes as the prices of its constituent stocks change.

3.3. Types of Basic Derivatives

- There are various types of derivatives traded on exchanges across the world.
- They range from the very simple to the most complex products.
- The following are the three basic forms of derivatives, which are the building blocks for many complex derivatives instruments :

Forwards

Futures

Options

- Knowledge of these instruments is necessary in order to understand the basics of derivatives.
- We shall now discuss each of them in detail.

3.3.1. Forwards

- A forward contract or simply a forward is a contract between two parties to buy or sell an asset at a certain future date for a certain price that is pre-decided on the date of the contract.
- The future date is referred to as **expiry date** and the pre-decided price is referred to as **Forward Price**.
- It may be noted that Forwards are **private contracts** and their terms are determined by the parties involved.

3.3.2. Futures

- Like a forward contract, a futures contract is an agreement between two parties in which the buyer agrees to buy an underlying asset from the seller, at a future date at a price that is agreed upon today.
- However, unlike a forward contract, a futures contract is not a private transaction but gets traded on a recognized stock exchange.
- In addition, a futures contract is standardized by the exchange.
- All the terms, other than the price, are set by the stock exchange (rather than by individual parties as in the case of a forward contract).

Future....

- In addition, both buyer and seller of the futures contracts are protected against the counter party risk by an entity called the **Clearing Corporation**.
- In case **one of the parties defaults**, the Clearing Corporation steps in to fulfill the obligation of this party, so that the **other party does not suffer due to non-fulfillment of the contract**.
- To be able to guarantee the fulfillment of the obligations under the contract, the Clearing Corporation **holds an amount as a security from both the parties**.
- This amount is called the **Margin money** and can be in the form of **cash or other financial assets**.

Difference between forwards and futures

Forwards	Futures
Privately negotiated contracts(traded over the counter)	Traded on an exchange
Not standardized	Standardized contracts
Settlement dates can be set by the parties	Fixed settlement dates as declared by the exchange
High counter party risk	Almost no counter party risk

3.3.3. Options ...

- Like forwards and futures, options are derivative instruments that provide *the opportunity to buy or sell an underlying asset on a future date*.
- An **option is a derivative contract between a buyer and a seller**, where one party (say First Party) gives to the other (say Second Party) the right, but not the obligation, to buy from (or sell to) the First Party *the underlying asset on or before a specific day at an agreed-upon price*.
- In return for granting the option, **the party granting the option collects a payment from the other party**.
- This payment collected is called the “**premium**” or **price of the option**.

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- **Call option** : the buyer has a rights but not an obligation to buy underlying asset
 - **Put option** : the seller has a rights but not an obligation to sell underlying asset

Option...

- The right to buy or sell is held by the “option buyer” (also called the option holder); the party granting the right is the “option seller” or “option writer”.
- Unlike forwards and futures contracts, options require a cash payment (called the premium) upfront from the option buyer to the option seller.
- Options can be traded either on the stock exchange or in over the counter (OTC) markets.
- Options traded on the exchanges are backed by the Clearing Corporation thereby minimizing the risk arising due to default by the counter parties involved.
- Options traded in the OTC market however are not backed by the Clearing Corporation.

Types of options

- There are two types of options—call options and put options.
- ***Call option***
- A call option is an option granting the *right to the buyer of the option to buy the underlying* asset on a specific day at an agreed upon price, *but not the obligation to do so*.
- *It is the seller* who grants this right to the buyer of the option.
- It may be noted that the person who has the **right to buy the underlying asset is known as the “buyer of the call option”**.

Call Option...

- Since the buyer of the call option has the right (but no obligation) to buy the underlying asset, he will **exercise his right to buy the underlying asset if and only if the price of the underlying asset in the market is more than the strike price on or before the expiry date of the contract.**
- The buyer of the call option does not have an obligation to buy if he does not want to.

- *Put option*

- A put option is an option granting to the owner the right, but not the obligation , to sell an asset at specified price (strike price), by predetermined date to a given party .

Difference Between Futures and Options

Futures	Options
Both the buyer and the seller are under an obligation to fulfill the contract.	The buyer of the option has the right and not an obligation whereas the seller is under obligation to fulfill the contract if and when the buyer exercises his right.
The buyer and the seller are subject to unlimited risk of loss.	The seller is subjected to unlimited risk of losing whereas the buyer has limited potential to lose (i.e. the option premium).
The buyer and the seller have potential to make unlimited gain.	The buyer has potential to make unlimited gain while the seller has a potential to make limited gain. On the other hand the buyer has a limited loss potential and the seller has unlimited loss potential.



END OF CHAPTER 3

THANK YOU !!!!!



UNIT FOUR

4. FINANCIAL INSTITUTION

4.1 Definitions of financial institution

- A financial institution (FI) is a company engaged in the business of dealing with financial and monetary transactions such as deposits, loans, investments, and currency exchange.

Cont....

- **Financial institutions** encompass a broad range of business operations within the financial services sector including banks, trust companies, insurance companies, brokerage firms, and investment dealers.
- Virtually everyone living in a developed economy has an **ongoing or at least periodic need** for the services of financial institutions.

4.2 Importance of Financial Institution

- **a. Motivating the Financial Sector**
- Generally Financial Institutions will only use their resources for the benefit of their **interest** - i.e. help to make profits, either directly or indirectly. The considerations are important because with the help of **development of institutions there is rise in the investment business in the country.**
- With presence of more institutions there will be **motivation** in the **financial/business area** to perform better and take steps for the strengthening of country.
- This will lead towards the **prosperity** in the country by removing the risk.

b. Financing the Small Scale Sector

- **Credit** is the key input for sustained growth of small scale sector and its availability is thus a matter of great importance.
- The provision of **short term credit or working capital to small businesses** for its day to day requirement for purchasing raw material and other inputs like water, electricity, etc. and

- 
- For payment of salaries and wages; and long term credit for creation of fixed assets like building, land, plant and machinery help the SME sector to perform better

C. Development and Support Services

- Financial institutions provides grants and loans for different organizations development and support to different agencies working for the promotion and development of industries like associations, chambers are available

D. Micro Finance Credit

- Micro Credit is offered to the poorest sector of country.
- This active step to facilitate growth of the micro finance sector in country is very commendable.
- It is predicted to emerge as the apex community by providing a broad range of financial and non-financial services such as grant support, loan funds, equity and institution building support

E. Trade Facilitation Programme

- Trade Facilitation Programme aims to foster trade in the countries of operations, both intra-regional and global.
- **Through the programme**, institutions provide promises to confirming banks, taking the political and commercial payment risk of international trade transactions take on by banks in the countries of operations.
- This pioneering programme remains an important source of trade finance in many of countries of operations.

F. Availability of Financial services to households & individuals

- Financial institutions can have a major effect on the activities of individuals by the provision of appropriate financial arrangements - for example, access to cheap mortgage finance is a requirement of widespread home ownership, and car ownership has been critically increased by the accessibility of car loans and hire purchase.
- In the absence of suitable financing arrangements, products may struggle to achieve sales, mainly if they have high capital costs.

G. Capital mobilization

- Capital mobilization is commonly one of the most **necessary conditions for economic development**.
- **The role played by Financial Institutions** in the process of financial integration in developing countries is very important.
- With the help of this channel advantage of integration materialized.
- **With the help of capital mobilization** capacity building, good governance and economic reforms can **easily be achieved**

H. Achievement of Growth

- Developed financial systems allow economies to reach their potential since they allow companies which have successfully recognized profitable opportunities to exploit these opportunities as intermediaries by channeling investment funds from those in the economy who are prepared to defer their consumption plans into the future.
- Achievement of development in country becomes easy with introduction of financial institutions.
- Different steps of financial development require adequate institutional procedures to be in place

I. Financial Innovation

- Development of Financial Institutions helps in **concentrating** on the improvements in technology and its influence on how financial products are delivered.

4.3 Types of Financial Institution

□ Broadly speaking, there are three major types of financial institutions.

1. **Depository institutions**

2. **Contractual institutions**

Let us see one by one

4.3.1 Depository institutions

- Deposit-taking institutions that accept and manage deposits and make loans, including :
 - A. **Commercial banks**
 - B. **Credit unions**
 - C. **Credit and saving association**
 - D. **Mutual Saving Banks**

A. Commercial Banks

- Commercial banks are those financial institutions, which help **in pooling** the savings of surplus units and arrange their productive uses.
- They basically accept the deposits from individuals and institutions, which are repayable on demand. These deposits from individuals and institutions are invested to satisfy the short-term financing requirement of business and industry.
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Cont...



- Deposit provided by commercial banks are commonly classified in three. Such as
 1. Current deposit
 2. Saving deposit
 3. Certificate deposit

B. Credit unions

- Credit unions **are cooperative associations** where large numbers of people are voluntarily associated for savings and borrowing purposes.
- These individuals are the members of credit unions as **they make share investment along with deposits**. The saving generated from these members are used to lend the members of the union only.
- The credit union should be controlled by its members.

C. Credit and saving association

- Saving and loan associations are the financial institutions involved in collecting funds of many small savers and lending these funds to home buyers and other types of borrowers.

D. Mutual Saving Banks

- Mutual saving banks are more or less similar to saving and loan associations.
- They primarily accept savings of individuals and they are lent to the **home users and consumers on a long-term basis.**

Advantages of depository institution

- they provide safekeeping services and liquidity
- they provide a payment system consisting of checks and electronic funds transfers;
- They pool the money of many savers and lend it out to people and businesses

4.3.2 Contractual institutions

- A. Insurance Companies**
- B. Pension Funds**
- C. Finance companies**
- D. Investment bank**
- E. Brokerage firm**

A. Insurance companies

- Insurance companies are the contractual saving institutions which collect **periodic premium** from insured party and in return agree to compensate against the risk of loss of life and properties.

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- Insurance companies may sell any combination of insurance types, but are often **classified into three groups**
 - 1. **Life insurance companies**, which sell life insurance, annuities and pensions products and bear similarities to asset management businesses
 - 2. **Non-life or property/casualty** insurance companies, which sell other types of insurance.
 - 3. **Health insurance companies**, which sometimes sell life insurance or employee benefits as well

B. Pension funds

- **Pension** ("payment") is a fund into which a sum of money is added during an employee's employment years, and from which payments are drawn to support the person's retirement from work in the form of periodic payments.
- A pension may be a **"defined benefit plan"** where a fixed sum is paid regularly to a person.

C. Finance companies

- Finance companies are the financial institutions that engage in satisfying individual credit needs, and perform merchant banking functions.
- In other words, finance companies are non-bank financial institutions that tend to meet various kinds of consumer credit needs.
- They involve in **project financing, housing and other kind of real estate financing**

D. Investment banks

- An **Investment bank** is a financial services company or corporate division that engages in advisory-based financial transactions on behalf of individuals, corporations, and governments.

E. Brokerage Firms

- A **broker** is a person or firm who arranges transactions between a buyer and a seller for a commission when the deal is executed. A broker who also acts as a seller or as a buyer becomes a principal party to the deal.

Advantages of non-depository institution

- These institutions receive the public's money because they offer other services than just the payment of interest.
 - ▣ They can spread the financial risk of individuals over a large group, or
 - ▣ Provide investment services for greater returns or for a future income.



□ *End of chapter Four*